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Quality Assessment of Surface Current and Wave Measurements from the High-Frequency Radar System in the Gulf of Trieste

Section 1: Collecting and editing metadata from HFR GOT

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1 Introduction

Due to wind and buoyancy forcings the circulation processes in the ocean surface tend to be the strongest and most unstable. High frequency (HF) instruments that operate within the radio bands of the electromagnetic spectrum (3 MHz - 50 MHz) use backscattered signals (*Bragg scattering*) from the ocean surface to map surface currents, wind, and waves.

Long-term deployments of HF radars (HFR) facilitate the characterisation of coastal circulation patterns, including wind-driven and tidal currents, as well as buoyant flows associated with river discharge. These observations also contribute to broader studies of marine ecology and coastal processes.

The Adriatic Sea is one of the hotspots of climate change, particularly the northern Adriatic shelf, bounded to the south by the Ancona-Zadar line. Due to its shallow depths, this area is exposed to large temperature fluctuations and strong currents, which are intermittently driven by the basin's dominant winds, the Bora and the Sirocco.

Because of these conditions, it is essential to monitor circulation in the northern Adriatic and in the Gulf of Trieste (GOT) over long, climate-relevant timescales.

To this end, a HFR system for measuring surface currents and waves was established in the GOT through cross-border cooperation between Slovenia and Italy.

This document evaluates metadata on the configuration of the High Frequency Radar - Northern Adriatic (HFR-NAdr) system for the period 2015–2026. It covers station locations, operating frequencies, temporal and spatial resolution, measured variables (currents, waves), data availability, and quality control (QC) algorithms.

Information was gathered from publicly available sources (websites, ERDDAP metadata, published articles) and supplemented with operator-confirmed values where available.

This data was collected, processed, and made freely available by The Regional Environmental Protection Agency of Friuli Venezia Giulia (ARPA FVG), The Environmental Agency of the Republic of Slovenia (ARSO), The National Institute of Biology (NIB), and The National Institute of Oceanography and Applied Geophysics (OGS), and the programs that contribute to it.

2 Network Overview

The NAdr network is a Wellen Radar (WERA) HF Radar System, which is an example of a beam-forming (BF) radar. The network comprises four antennas located across Italy and Slovenia: AURI, TRI1, IZOL, and PIRA.

Figure 2 illustrates the study area and the locations of the four stations, while Table 1 provides an overview of the system with information on the operating frequencies and the spatial and temporal resolutions.

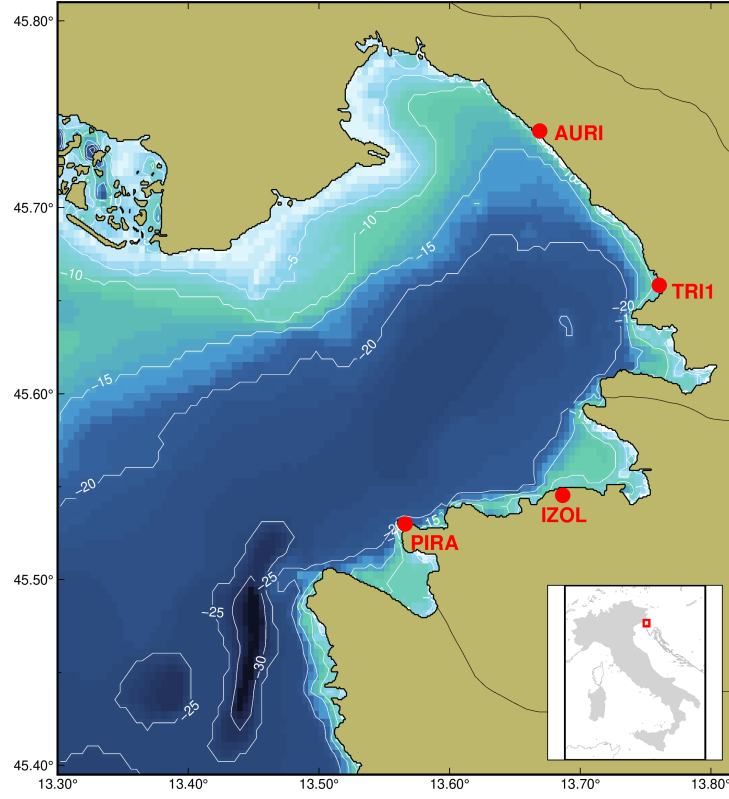


Figure 2: Bathymetry (GEBCO) of the Gulf of Trieste (the contour interval is 5 m). The location of the four HFR stations, AURI, TRI1, IZOL, and PIRA, are marked by red dots.

System Name	HFR - NAdr
Area	Northern Adriatic Sea - Gulf of Trieste
Operators	ARPA, ARSO, OGS, NIB
Radar System	WERA HFR System
Operating Frequency	24.5 - 24.635 MHz
Spatial resolution	1.5 km grid (0.0135 ° in latitude; 0.0193° in longitude)
Temporal Resolution	30 minutes
Radial Coverage	35 - 40 km
Depth represented	upper 0.3 - 2.5 m
Time Period	(2015) 2021 - present
DOI	https://doi.org/10.57762/8RRE-0Z07

Table 1: System overview of the HFR-NAdr.

3 Funding Projects

- TOSCA (Tracking Oil Spills and Coastal Awareness): HF radar experiments in the Gulf of Trieste (2012 - 2014)
- HAZADR (Strengthening common reaction capacity to fight sea pollution): Installation of WERA stations AURI and PIRA (2012 - 2015)
- Firespill (Interreg IT-HR): Installation of TRI1 station in Trieste (2021 - 2023)
- SHAREMED: Installation of IZOL station in Izola (2022)

4 Stations

Station	AURI	TRI1	IZOL	PIRA
Operator	OGS	ARPA FVG	NIB	ARSO
Location	Aurisina, Italy	Trieste, Italy	Izola, Slovenia	Piran, Slovenia
Coordinates	13.669°E 45.741°N	13.7605°E 45.6583°N	13.6866°E 45.5454°N	13.566°E 45.53°N
Frequency [MHz]	24.635	24.525	24.5	24.525
Operational Since	2015	10/07/2023	01/03/2023	2015
Status	Active	Active	Active	Active

Table 2: Technical specifications of the four HFR-NAdr network stations.

4.1 Network configuration by period

Period	Active Stations
2014	AURI, PIRA
2015	AURI only (PIRA down due to storm damage)
March 2023 - present	AURI, PIRA, IZOL
July 2023 - present	AURI, PIRA, TRI1, IZOL

Table 3: Network configuration by period for the HFR-NAdr.

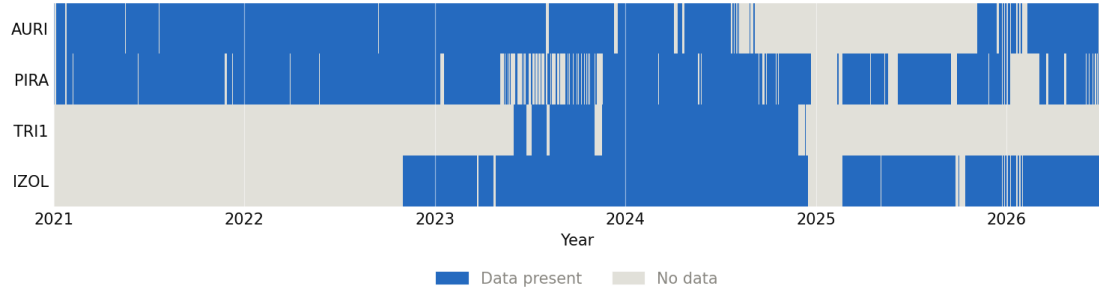


Figure 3: Measured hourly data availability per station, 2021–present, based on the radial current archives downloaded from the EU HFR Node. Blue indicates at least one valid measurement in that hour; gray indicates no data. Station start dates match Table 3. A shared gap of roughly ten months (late 2024 to late 2025) affects all stations and reflects an interruption in the archive itself rather than a per-station outage.

5 Data Availability

Public data sources

Source	URL	Content	Period
EU HFR Node ERDDAP	erddap.hfrnode.eu	Radials (4 stations) + Totals (v3)	2021–present
EMODnet Physics ERDDAP	erddap.emodnet-physics.eu	Totals (HFRADAR_NADR_Totals)	2021–present
OGS THREDDS	dsecho.ogs.it/thredds/ ¹	Merged datasets	2021–2024
ARPA FVG	meteo.fvg.it/radar_marino.php	Currents + waves (visualization & archive)	–

Table 4: Overview of Data Repositories, Available Products, and Temporal Coverage.

5.1 ERDDAP datasets (EU HFR Node)

Note: No wave datasets for HFR-NAdr are present on ERDDAP (search confirmed 2026-06-24).

Dataset ID	Description
EUHFR_NRTcurrent_HFR-NAdr-AURI_v3	Radial velocities, AURI station
EUHFR_NRTcurrent_HFR-NAdr-TRI1_v3	Radial velocities, TRI1 station
EUHFR_NRTcurrent_HFR-NAdr-IZOL_v3	Radial velocities, IZOL station
EUHFR_NRTcurrent_HFR-NAdr-PIRA_v3	Radial velocities, PIRA station
EUHFR_NRTcurrent_HFR-NAdr-Total_v3	Total surface velocities
EUHFR_NRTcurrent_HFR-NAdr-Total_v3_ Last4Months_QC	Total velocities, last 4 months only

Table 5: Erddap datasets description.

6 Measured Variables

6.1 Surface currents

Each station measures radial current velocities toward/away from the antenna. Combining radials from two or more stations yields *total surface current vectors* on a regular spatial grid.

6.1.1 Radial Velocity

The dataset consists of maps of radial velocity of the sea water surface current collected in the GOT. Data are averaged over a time interval of 30 minutes. HFR measurements of ocean velocity are radial in direction relative to the radar locations and representative of the upper 0.3-2.5 meters of the ocean.

Table 6 provides an overview of the variables included in the dataset.

6.1.2 Total Velocity

The data set consists of maps of total velocity of the surface current in the GOT averaged over a time interval of 30 minutes. Surface ocean velocities estimated by HFR are representative of the upper 0.3-2.5 meters of the ocean.

Total velocities are derived using least square fit that maps radial velocities measured from individual sites onto a cartesian grid. The final product is a map of the horizontal components of the ocean currents on a regular grid in the area of overlap of two or more radar stations.

Table 7 provides an overview of the variables included in the dataset.

6.2 Waves

WERA radars measure wave parameters from the second order Doppler spectrum.

Wave data are not available on any public ERDDAP or THREDDS server. The EU HFR Node and EMODnet Physics publish currents only.

Variable	Description	Unit
DRVA	Direction of radial vector away from instrument	degree
EACC	Radial accuracy of current velocity over coverage period	$\frac{m}{s}$
EWCT	Surface eastward sea water velocity	$\frac{m}{s}$
HCSS	Radial variance of current velocity over coverage period	$\frac{m^2}{s^2}$
NSCT	Surface northward sea water velocity	$\frac{m}{s}$
RDVA	Radial sea water velocity away from instrument	$\frac{m}{s}$
		Threshold
AVRB_QC	Average radial bearing quality flag	bearing range = $0.0 - 360.0^\circ$
CSPD_QC	Velocity threshold quality flag	max. velocity = $1.2 \frac{m}{s}$
MDFL_QC	Median filter quality flag	distance limit = 5.0 km; velocity median difference threshold = $1.0 \frac{m}{s}$
OWTR_QC	Over-water quality flag	land/sea mask (GeoPandas naturalearth_lowres); not a numeric threshold
POSITION_QC	Position quality flag	no fixed numeric threshold (coordinate check/composite of the above)
QCflag	Overall quality flag	no fixed numeric threshold (coordinate check/composite of the above)
RDCT_QC	Radial count quality flag	minimum number of radial vectors = 25
VART_QC	Variance threshold quality flag	maximum variance = $1.0 \frac{m^2}{s^2}$

Table 6: Grid Variables for Near Real Time Surface Ocean Radial Velocity by HFR-NAdr.

Variable	Description	Unit
EWCT	Surface eastward sea water velocity	$\frac{m}{s}$
GDOP	Geometrical dilution of precision	
NSCT	Surface northward sea water velocity	$\frac{m}{s}$
UACC	Accuracy of surface eastward sea water velocity	$\frac{m}{s}$
VACC	Accuracy of surface northward sea water velocity	$\frac{m}{s}$
		Threshold
CSPD_QC	Velocity threshold quality flag	maximum velocity = $1.0\frac{m}{s}$
DDNS_QC	Data density threshold quality flag	minimum contributing radial velocities = 3
GDOP_QC	GDOP threshold quality flag	maximum GDOP = 2.0
POSITION_QC	Position quality flag	same as above (Table 6)
QCflag	Overall quality flag	same as above (Table 6)
VART_QC	Variance threshold quality flag	maximum variance = $1.0\frac{m^2}{s^2}$

Table 7: Grid Variables for Near Real Time Surface Ocean Total Velocity by HFR-NAdr-Total.

7 Key Publications

1. **Chiggiato et al. (2025)** — "Influence of wind stress and the Isonzo/Soča River outflow on surface currents in the Gulf of Trieste", *Ocean Science*, 21, 2197–2214. — Confirms 4 stations, wave measurements (Hs, direction), data period Oct–Nov 2023.
2. **Révelard et al. (2023)** — "Superstatistical analysis of sea surface currents in the Gulf of Trieste", *NPG*, 30, 515–531. — Analysis of data Jan 2021 – Oct 2022 (2 stations: AURI, PIRA). Documents transfer of PIRA management from NIB to ARSO in July 2022.
3. **Menna et al. (2022)** — "Coastal high-frequency radars in the Mediterranean – Part 2", *Ocean Science*, 18, 797–837. — Overview of HFR networks in the Mediterranean, including NAdr.
4. **Berta et al. (2021)** — "Multi-Platform, High-Resolution Study of a Complex Coastal System: The TOSCA Experiment in the Gulf of Trieste", *J. Mar. Sci. Eng.*, 9(5), 469. — Description of the 2012 TOSCA experiments.

8 Information Sources

Source	Link
EU HFR Node — HFR-NAdr network	https://www.hfrnode.eu/networks/hfr-nadr-2/
ERDDAP — EU HFR Node	https://erddap.hfrnode.eu/erddap/search/index.html?searchFor=NAdr
EMODnet Physics ERDDAP — NAdr Totals	https://erddap.emodnet-physics.eu/erddap/info/HFRADAR_NADR_Totals/index.html
OGS THREDDS — NAdr-radar	https://dsecho.ogs.it/thredds/catalog/radar/NAdr-radar/catalog.html
ARPA FVG — Firespill news	https://www.arpa.fvg.it/temi/progetti/progetti-europei/news/nuovo-radar-marino-per-i-monitoraggi-nel-golfo-di-trieste-grazie-al-progetto-firespill/
Chiggiato et al. (2025), Ocean Science	https://os.copernicus.org/articles/21/2197/2025/
Révelard et al. (2023), NPG	https://npg.copernicus.org/articles/30/515/2023/
NIB — HF radar (SHAREMED/IZOL installation)	https://www.nib.si/mbp/en/oceanographic-data-and-measurements/other-oceanographic-data/hf-radar-2
SHAREMED — official project site	https://sharemed.interreg-med.eu/

Table 8: Relevant sources and data repositories for the HFR-NAdr network.